

# Chemical Identity Lost in Regulation: A Study of Semantic Interoperability in European Chemical Substance Data

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## Abstract

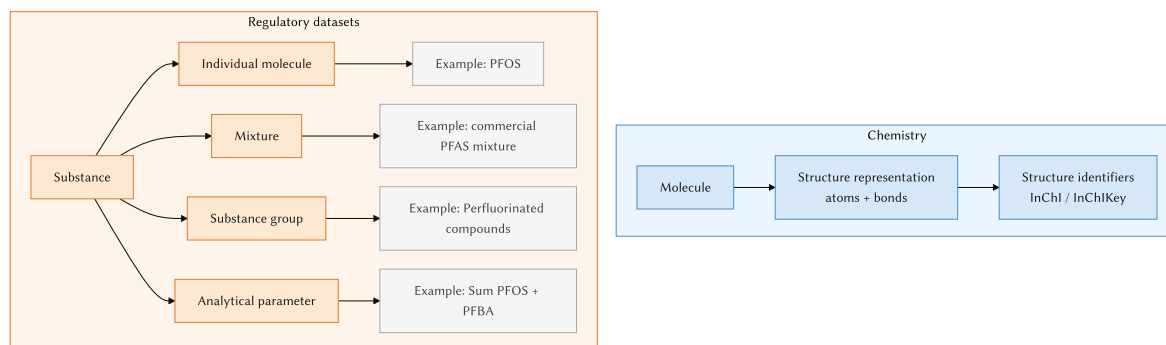
European regulatory datasets represent chemical substances heterogeneously: CAS numbers, EC numbers, and textual names that may refer to single molecules, mixtures, substance groups, or analytical parameters. We analyse 18 ECHA-based datasets (41,813 records) using a seven-tier linkability taxonomy. Only 62.5% of entries link to a defined molecular structure; 37.5% are non-structure-based. CAS numbers prove unreliable as unique identifiers. Four sentence-embedding models achieve ChemOnt Hit@1  $\leq 5.75\%$ ; Claude Sonnet 4.6 reaches 39.2% yet still misclassifies the majority. Structure-defined substances are integrated into a knowledge graph with ChemOnt classification and ChEBI biological roles (~48,400 cross-domain links). For non-structure entries, the LLM abstains from direct classification far more often than for structure-defined substances, appropriately reflecting their lack of a defined structure; a separate LLM-based scope-mapping assessment yields 304 validated SKOS triples linking regulatory group entries to ChemOnt nodes. Semantic interoperability requires structural changes at the level of regulatory data modelling and legislation.

## Keywords

semantic interoperability, regulatory data, chemical substance identification, knowledge graph, REACH

## 1. Introduction and Related Work

In chemistry, substance identity is operationalised through structure-based identifiers such as the International Chemical Identifier (InChI) and InChIKey [1], enabling unambiguous linking across databases such as PubChem [2] and the Chemical Entities of Biological Interest database (ChEBI) [3]. In European regulation, defined under the Registration, Evaluation, Authorisation and Restriction of Chemicals framework (REACH) [4] and the Classification, Labelling and Packaging regulation (CLP) [5], a *substance* may refer to a single molecule, a mixture of unknown or variable composition (UVCB) [6], a group of related compounds (e.g. per- and polyfluoroalkyl substances, PFAS, which encompasses individual molecules such as PFOS without explicit set-membership semantics; Figure 2), or an analytical sum parameter, as summarised in Figure 1.



**Figure 1:** Conceptual contrast between chemistry (structure-based identity via InChI/InChIKey) and regulatory datasets (heterogeneous substance types).

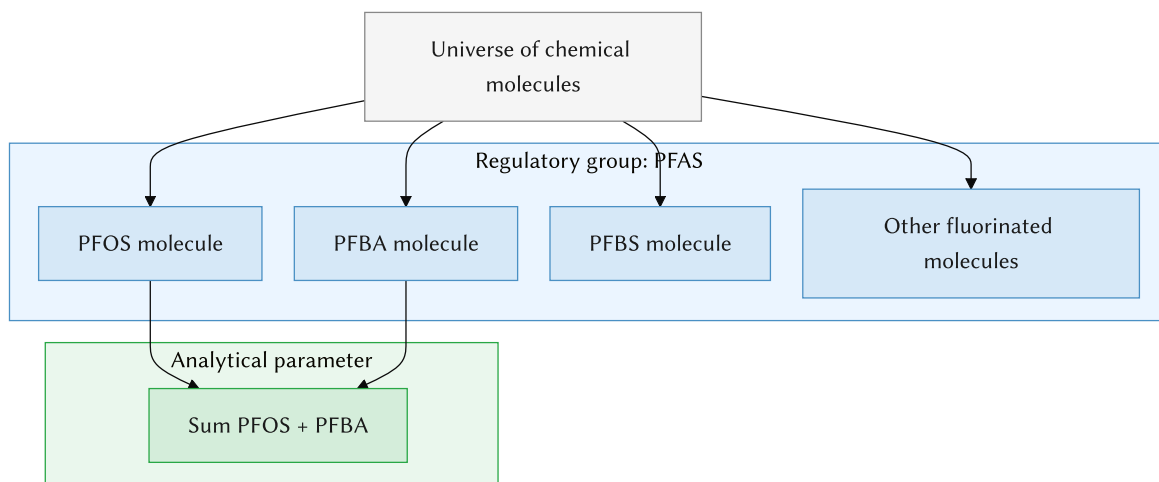
Posters, Demos, Blue Sky, and Tutorials at SEMANTiCS 2026, Sep 2026, Ghent, Belgium

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**Figure 2:** PFAS as a regulatory group without explicit set-membership semantics; *Sum PFOS + PFBA* as a derived analytical parameter.

Such regulatory entities (whether single molecules, UVCBs, substance groups, or sum parameters) are identified by Chemical Abstracts Service (CAS) numbers [7], textual names, or both—identifiers that carry no structural information. Much of this information remains confidential or inaccessible in machine-readable form: REACH restricts public disclosure of the full Substance Identity Profile and UVCB composition [8], leaving only non-machine-readable safety data sheets and classifications publicly available [9, 10]. This semantic mismatch limits automated integration [11] and FAIR (Findable, Accessible, Interoperable, Reusable) data reuse [12]. Prior work shows that structure-based identifiers and ontology-driven knowledge graphs enable interoperability in chemistry and in single-domain regulatory contexts such as cosmetics risk assessment [13], while legal-data integration across jurisdictions remains fragmented [14]. However, the cross-dataset mismatch between structure-based chemical identity and regulatory substance definitions has not previously been systematically characterised.

We address three research questions: (RQ1) To what extent can regulatory substances be linked to a unique chemical structure? (RQ2) How reliable are CAS numbers as cross-dataset identifiers? (RQ3) Can embedding-based or LLM methods compensate for absent structure identifiers? Contributions: (C1) a seven-tier linkability taxonomy; (C2) empirical demonstration of CAS inconsistency; (C3) a benchmark showing the failure of semantic approaches; (C4) an integrated knowledge graph linking 18 sources with ChemOnt [15] and ChEBI; (C5) 304 validated SKOS (Simple Knowledge Organization System) [16] mapping triples as a legislative–ChemOnt crosswalk.

## 2. Methods

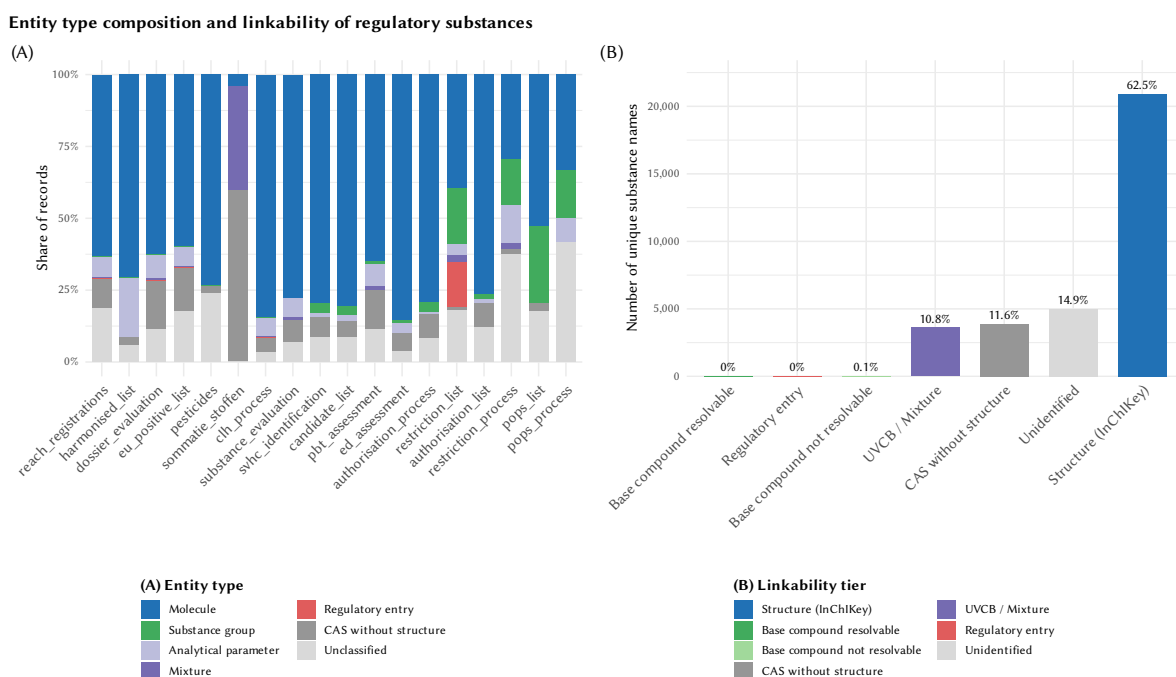
**Data integration and linkability.** We integrated 18 sources: six European Chemicals Agency (ECHA) obligation lists (Candidate [17], Authorisation, Restriction, POPs, EU Positive, CLH), nine ECHA activity lists, REACH registrations [18], the EU Pesticides Database, OSPAR priority chemicals, and a Flemish administrative substance list. Per entry, CAS, EC number [19], and substance name were collected; substances were linked to InChIKey via PubChem [2]. Each entry was then assigned to one of seven tiers: tier 1 (direct InChIKey link), tiers 2–3 (partially resolvable), tier 4 (unresolved CAS), tier 5 (mixture or UVCB [6]), tier 6 (administrative entry), tier 7 (no usable identifier). InChI extensions such as MInChI [1] support representation of some mixtures in principle, but are absent from current European regulatory datasets; tier 5 entries therefore resist structure-based linking. Bidirectional CAS–InChIKey mappings were analysed to assess identifier consistency [20].

**Semantic matching.** We test the hypothesis that semantic similarity can serve as a proxy for chemical taxonomy assignment, expecting it to fail because substance names for groups and mixtures carry no

structural information. Substance names and ~3,800 ChemOnt class labels [15] were encoded using four sentence-transformer models [21] (MiniLM, MPNet, SciBERT, BioBERT). Performance was measured using Hit@*k* against 28,204 ClassyFire-assigned, tier-1 (structure-linked) substances as ground truth, since only these carry a verifiable ChemOnt class. Given the embedding models’ near-complete failure on this benchmark (Table 1), they were not carried forward to the harder, tier 4/5/7 non-structure population. Claude Sonnet 4.6 was applied to the same tier-1 benchmark under zero-shot prompting, then, for the tier 4/5/7 non-structure entries, further used to (a) classify ChemOnt membership directly and (b) assess how the entry’s name relates to a ChemOnt node (exact, subset, or broader scope), using SKOS mapping types [16]. Structure-defined substances were integrated into an RDF knowledge graph with ChEBI biological roles [3], and a ToxPi-inspired composite priority score [22] was computed to illustrate the analytical value.

### 3. Results

**Linkability (RQ1).** Integration of 18 sources yielded 41,813 records. Applying the seven-tier taxonomy to 33,452 unique entries (Figure 3) shows that 62.5% (20,909) link directly via InChIKey (tier 1), while 37.5% cannot be mapped to a defined structure: 11.6% carry unresolved CAS numbers (tier 4), 10.8% are mixtures or UVCBs (tier 5), and 14.9% lack any usable identifier (tier 7). Tier composition differs substantially between regulatory instruments. Tiers 2–3 and 6 together cover under 0.3% of entries here; the taxonomy retains them to generalise to sources where these categories are populated.



**Figure 3:** (A) Entity type composition across 18 sources. (B) Seven-tier linkability taxonomy: 62.5% of 33,452 unique entries link directly via InChIKey (tier 1); 14.9% lack any usable identifier (tier 7).

**CAS number reliability (RQ2).** 70.0% of structure-resolved InChIKeys appear in only a single source, making CAS-based cross-dataset linking fundamentally insufficient as a shared key. Ambiguity compounds this: 195 InChIKeys are each associated with more than one CAS number (up to seven), while 4 CAS numbers resolve to two distinct InChIKeys [20, 23]; for example, CAS 26471-62-5 (toluene diisocyanate) maps to two positional isomers with different InChIKeys across sources. Even for structure-defined substances, then, a flat CAS number cannot substitute for a structure-based identifier: the same semantic gap that leaves non-structure entities unlinked entirely (RQ1).

**Failure of semantic approaches (RQ3).** Table 1 shows that, on the tier-1 (structure-linked) benchmark, embedding-based matching fails: the best model (all-MiniLM-L6-v2) achieves Hit@1 of 5.75%, meaning the correct ChemOnt class is retrieved in fewer than 1 in 17 attempts; all others score below 1%. Claude Sonnet 4.6 reaches Hit@1 = 39.2% (seven-fold improvement); high-confidence assignments reach 53.9% accuracy while low-confidence cases score 0%. The LLM still misclassifies the majority of substances. Applied to the tier 4/5/7 non-structure entries that make up the 37.5% unlinked population (RQ1), the LLM abstains from classification far more often than for structure-defined substances, confirming that textual descriptions cannot substitute for structural information.

**Table 1**

ChemOnt classification performance (direct-parent level). Ground truth: 28,204 ClassyFire-assigned substances [15].

| Model                          | Hit@1  | Hit@3  | Hit@5  | Hit@10 |
|--------------------------------|--------|--------|--------|--------|
| all-MiniLM-L6-v2               | 0.0575 | 0.0986 | 0.1234 | 0.1621 |
| all-mpnet-base-v2              | 0.0093 | 0.0152 | 0.0184 | 0.0247 |
| SciBERT                        | 0.0040 | 0.0071 | 0.0102 | 0.0131 |
| BioBERT                        | 0.0047 | 0.0077 | 0.0099 | 0.0150 |
| Claude Sonnet 4.6 <sup>†</sup> | 0.392  | —      | —      | —      |

<sup>†</sup>LLM: single label; only Hit@1 is comparable across rows.

**Knowledge graph and SKOS crosswalk (C4, C5).** Figure 4 summarises the integrated knowledge graph: ~17,500 regulatory substances (SKOS concepts) are linked to structure-based identity, chemical classification, and biological roles. This is fewer than the 20,909 tier-1-linked entries (RQ1) because names sharing one InChIKey collapse into a single concept. Structure-defined substances were integrated across lists, ChemOnt, and ChEBI, yielding ~48,400 cross-domain links (40,644 list memberships, 7,794 ChEBI roles) and enabling composite priority scoring [22] infeasible on fragmented datasets.

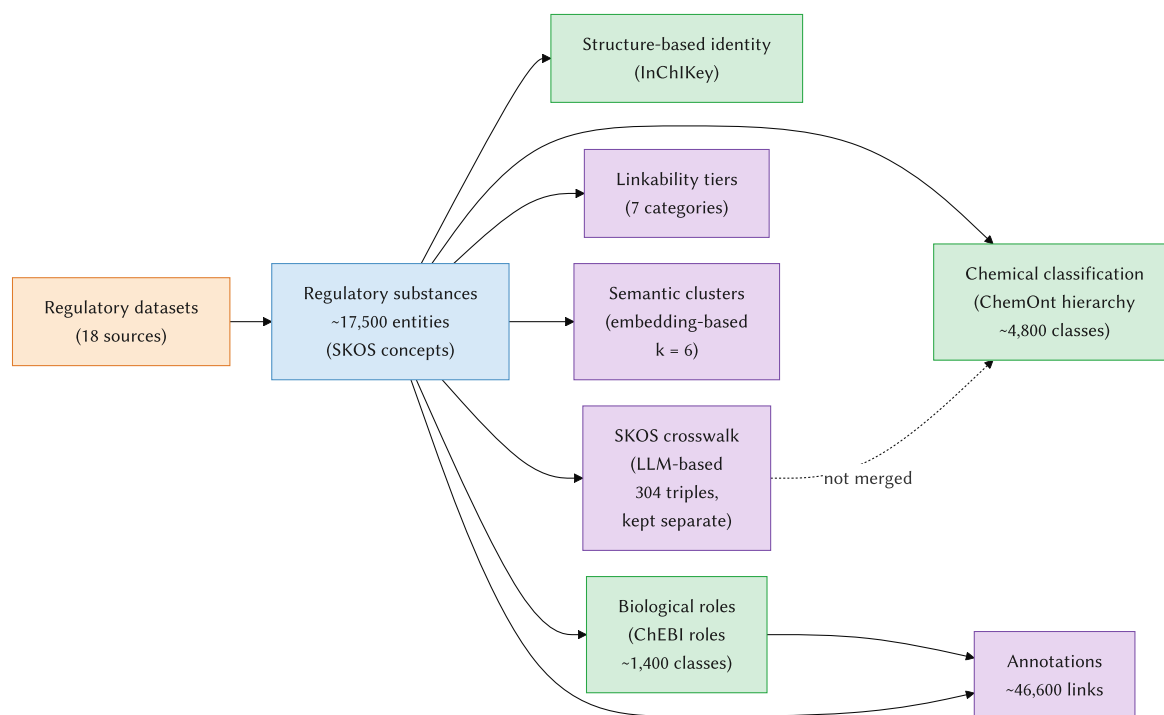
LLM-based scope-mapping identifies 31.7% of sampled non-structure entries as mappable to a ChemOnt node (vs. 0.8% by pattern matching), yielding 304 high-confidence SKOS triples (285 broadMatch, 18 exactMatch, 1 narrowMatch). The dominance of broadMatch reflects that regulatory group entries typically define constrained subsets of ChemOnt classes (e.g. *fatty acids*, *C16–22*  $\subset$  *Fatty acids and conjugates*), making scope machine-interpretable. The remaining 68% of non-structure entries could not be reliably mapped.

## 4. Discussion and Conclusion

Over one-third of regulatory substance entries cannot be linked to a molecular structure; CAS numbers introduce ambiguity rather than resolving it; and neither embeddings nor LLMs can substitute for structural information. Full manual reconciliation of non-structure entries (an estimated 3,150 person-years for REACH alone<sup>1</sup>) is infeasible.

These findings show that downstream harmonisation cannot substitute for change at the source: the different regulatory instruments that define and collect substance data (REACH, CLP, and the sector-specific lists built on them) must encode structure-based and group-level semantics consistently, not only the systems that integrate them afterwards. Three recommendations follow: (1) require structure-based identifiers (InChIKey; MInChI [1] for mixtures where applicable) across regulatory instruments; (2) mandate explicit representation of substance groups using formal ontologies such as ChemOnt; (3) publish regulatory data as linked data [24] with persistent URIs and SKOS taxonomy links, enabling FAIR-compliant [12] integration under the Interoperable Europe Act [11]. Future work should assess generality beyond ECHA/EU data (e.g. US EPA, UK HSE) and develop curated resolution

<sup>1</sup>Worst-case pairwise reconciliation (comparing every pair; real reconciliation would cluster) of  $n \approx 11,250$  non-structure entries needs  $n(n-1)/2 \approx 6.3 \times 10^7$  comparisons; at ~100/day and 200 working days/year, that is  $\approx 3,150$  person-years.



**Figure 4:** Overview of the integrated knowledge graph: regulatory substances (SKOS concepts) are linked to structure-based identity (InChIKey), chemical classification (ChemOnt), and biological roles (ChEBI), with derived linkability tiers, semantic clusters, and an LLM-based SKOS crosswalk kept separate from (not merged into) the ChemOnt hierarchy.

workflows for the 37.5% of entries that resist automated linking, guided by the taxonomy introduced here.

*Limitations:* Scoped to 18 ECHA-based datasets; ClassyFire ground truth is itself algorithmically assigned, not expert-validated, and a manual sample shows most apparent LLM errors select a chemically valid alternative label rather than a hallucinated one; LLM evaluation uses a single commercial model (Claude Sonnet 4.6); 68% of non-structure entries resist automated alignment. *Code and data:* <https://doi.org/10.5281/zenodo.21512303> [25]. *AI declaration:* Claude Sonnet 4.6 was used as the object of study for the LLM experiments (~\$100 API) and as a development tool. All content was reviewed by the authors, who take full responsibility for the publication’s content.

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